

Quantum Mechanics Problem Sheet 1

Solutions, with reference to my computational models

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Introduction

These are my solutions to Quantum Mechanics Problem Sheet 1. Several of the questions ask for quantities that the models I built for the challenge tasks already calculate, so where that happens I have solved the problem by hand first and then checked it against the model. Those checks are in the boxes marked *Model check*.

Question	Topic	Related model
Q3	Stefan's law and Wien's law	Task 3, Planck spectrum
Q4	Photoelectric effect	Task 4, stopping voltage
Q5	de Broglie waves and Bragg scattering	Task 6, electron diffraction
Q6	Hydrogen spectral series	Task 5, Bohr model

Constants used throughout: $\sigma = 5.67 \times 10^{-8} \text{ W}/(\text{m}^2 \text{ K}^4)$, $h = 6.63 \times 10^{-34} \text{ J s}$, $c = 2.998 \times 10^8 \text{ m/s}$, $e = 1.60 \times 10^{-19} \text{ C}$, $m_e = 9.109 \times 10^{-31} \text{ kg}$.

What the model checks are testing

A model that uses the same equation I solved by hand cannot really confirm anything, since rearranging an equation and getting back where you started is not evidence of much. So in each case I have tried to make the model reach the answer by a different route:

- **Q3.** I use Wien's law by hand; the model maximises the Planck function numerically. Wien's law is derived by differentiating that function, so agreement tests the implementation rather than restating a constant.
- **Q4.** Rather than reading off the threshold, which would be circular, I fit the plotted line and recover h and ϕ from its gradient and intercept.
- **Q5.** The hand calculation goes from angle to voltage; the model goes from voltage to angle. It also recovers the atomic spacing d from a fitted gradient, which is a full round trip.
- **Q6.** I use the 91.13 nm formula by hand; the model builds the level energies from μ , e , ϵ_0 and h and takes differences. The small disagreement is itself informative, as explained there.

1 Atomic and nuclear scales

(i) Radius of an atom and of a nucleus

An atom has a radius of order 10^{-10} m, which is where the Bohr radius $a_0 = 5.29 \times 10^{-11}$ m sits, and a nucleus has a radius of order 10^{-15} m. So

$$\frac{r_{\text{atom}}}{r_{\text{nucleus}}} \sim \frac{10^{-10}}{10^{-15}} = 10^5.$$

The atom is about a hundred thousand times wider than its nucleus. In volume, which is what actually matters for how empty the atom is, that becomes a factor of 10^{15} .

The usual comparison is a pea in the middle of a football stadium, and it is worth taking seriously: essentially all of the mass sits in a region occupying 10^{-15} of the space, and the rest is electron cloud. It is also the reason Rutherford's result in Question 2 was such a surprise, and the reason solids feel solid at all is electrostatic repulsion between electron clouds rather than anything physically filling the volume.

(ii) Atoms in a marble, marbles in the Earth

The marble has radius 1.8×10^{-2} m, so

$$V_{\text{marble}} = \frac{4}{3}\pi(1.8 \times 10^{-2})^3 = 2.44 \times 10^{-5} \text{ m}^3.$$

Taking each atom to occupy a cube of side 2×10^{-10} m, one atomic diameter across, gives $V_{\text{atom}} \approx 8 \times 10^{-30} \text{ m}^3$ and

$$N_{\text{atoms}} = \frac{2.44 \times 10^{-5}}{8 \times 10^{-30}} \approx 3 \times 10^{24}.$$

The Earth has radius 6.4×10^6 m, so

$$V_{\text{Earth}} = \frac{4}{3}\pi(6.4 \times 10^6)^3 = 1.10 \times 10^{21} \text{ m}^3, \quad N_{\text{marbles}} = \frac{1.10 \times 10^{21}}{2.44 \times 10^{-5}} \approx 4.5 \times 10^{25}.$$

So there are of order 10^{24} atoms in the marble and of order 10^{25} marbles in an Earth-sized container. The two numbers are within about a factor of ten of each other, which is the point of the question: a marble sits roughly halfway between an atom and the Earth on a logarithmic scale of size.

Two things I have glossed over, both deliberately. Spheres cannot fill space, so the marble count should carry a packing fraction of about 0.74, bringing it down to 3.3×10^{25} ; and treating atoms as cubes rather than spheres was a way of packing them without needing the same correction twice. Since both counts are volume ratios, a packing correction applied to both would largely cancel in the comparison, which is why an order of magnitude estimate survives the sloppiness. I have also ignored that the marble is glass, a mixture of elements with different atomic radii, and that the Earth is not an empty container.

2 The Rutherford experiment

Geiger and Marsden fired alpha particles from a radioactive source at a thin gold foil, a few thousand atoms thick, and recorded where they went using a movable zinc sulphide screen that produced a small flash of light wherever an alpha particle struck it. The screen could be swung round to any angle, including back towards the source.

Almost all of the alpha particles passed straight through with little or no deflection. A small fraction were deflected through large angles, and about one in eight thousand came back towards the source, scattering through more than 90° . Rutherford's remark was that it was as though a shell fired at tissue paper had bounced back.

The prevailing model at the time was Thomson's, in which the positive charge was spread evenly throughout the atom with the electrons embedded in it. A diffuse charge distribution like that produces only weak fields, so it can deflect an alpha particle by a fraction of a degree but cannot turn one round. Backscattering is not a small quantitative disagreement with the model; it is something the model forbids.

Three conclusions follow. Most of the atom is empty, since most alphas pass through undeviated. The positive charge is concentrated in a very small volume, because only a large charge at close range produces a field strong enough to reverse an alpha particle, and the rarity of those events puts an upper bound on the size of the target. And essentially all of the mass is in that same small volume, because a light scatterer would recoil and carry away momentum rather than turning the alpha particle back.

Working from how the count rate fell off with angle, Rutherford put the nuclear radius at under 10^{-14} m, consistent with the 10^{-15} m used in Question 1 and smaller than the atom by the factor of 10^5 found there. This is the origin of the nuclear model of the atom, and the reason a Coulomb potential $\propto 1/r$ from a point-like nucleus is the right starting point for the hydrogen levels used in Question 6.

3 Stefan's law and Wien's law

(i) Temperature of a tungsten filament

The filament is a thin strip, so it radiates from both of its large faces and the total area is

$$A = 2 \times 0.20 \text{ m} \times 0.14 \times 10^{-3} \text{ m} = 5.60 \times 10^{-5} \text{ m}^2.$$

Rearranging Stefan's law $P = \sigma AT^4$ for T ,

$$T = \left(\frac{P}{\sigma A} \right)^{1/4} = \left(\frac{100}{5.67 \times 10^{-8} \times 5.60 \times 10^{-5}} \right)^{1/4} = 2369 \text{ K} = 2096^\circ \text{C}.$$

I have assumed the filament is a perfect black body, that all 100 W is radiated rather than partly lost by conduction through the supports, and that only the two flat faces emit.

This seems reasonable. Tungsten melts at about 3695 K, so 2369 K is hot enough to glow but safely below melting. A real filament has emissivity less than one, and since $T \propto \varepsilon^{-1/4}$ the true temperature would be a little higher.

(ii) Luminosity of the Sun

The Sun radiates from its whole surface area $4\pi R_\odot^2$, so

$$L = \sigma T^4 \cdot 4\pi R_\odot^2 = 5.67 \times 10^{-8} \times 5778^4 \times 4\pi \times (6.955 \times 10^8)^2 = 3.84 \times 10^{26} \text{ W},$$

which agrees with the accepted value of 3.83×10^{26} W.

(iii) Solar panel on Mars

By the time the light reaches Mars it is spread over a sphere of radius $d = 227.9 \times 10^9$ m, so the flux is

$$F = \frac{L}{4\pi d^2} = \frac{3.84 \times 10^{26}}{4\pi (227.9 \times 10^9)^2} = 589 \text{ W/m}^2.$$

At 20 % efficiency the panel needs area

$$A = \frac{P}{\eta F} = \frac{100}{0.20 \times 589} = 0.850 \text{ m}^2,$$

so its width is $\sqrt{0.850} = 0.92$ m.

As a check, the flux at Earth is about 1360 W/m^2 . Mars is 1.52 times further out, and $1360/1.52^2 = 589$, which matches.

(iv) Radiation from human skin

Skin at 34 °C is at $T = 307$ K, so Wien's law gives

$$\lambda_{\max} = \frac{2.90 \times 10^{-3}}{307} = 9.45 \times 10^{-6} \text{ m} = 9450 \text{ nm}.$$

This is in the infrared, well past the red end of the visible range, which is why body heat is invisible but shows up on a thermal camera. Core temperature at 37 °C gives 9360 nm, almost identical, since three kelvin on top of three hundred is only a one percent change.

(v) Spectra of three stars

For the Sun at 5778 K,

$$\lambda_{\max} = \frac{2.90 \times 10^{-3}}{5778} = 502 \text{ nm},$$

which is green, near the middle of the visible range. The Sun looks white rather than green because the peak is broad and all the visible wavelengths come out at similar intensities.

A cooler red star peaks at a longer wavelength with a lower curve, and a hotter blue star peaks at a shorter wavelength with a much higher one, since the total area under the curve goes as T^4 . The curves never cross, because a hotter body emits more at every wavelength.

Model check. My Task 3 model evaluates $B(\lambda, T) = \frac{2hc^2}{\lambda^5} (e^{hc/\lambda k_B T} - 1)^{-1}$ and finds its maximum numerically. At 5778 K it gives 501.4 nm, against 501.9 nm from Wien's law, a difference of 0.1 % which is just the resolution of my wavelength grid. This is a useful check because Wien's law comes from differentiating the Planck function, so agreement means my $B(\lambda, T)$ is right and not just roughly the correct shape.

(vi) Expansion to a red giant

A peak at 680 nm corresponds to

$$T_2 = \frac{2.90 \times 10^{-3}}{680 \times 10^{-9}} = 4265 \text{ K}.$$

The luminosity is unchanged, so $\sigma T_1^4 \cdot 4\pi R_1^2 = \sigma T_2^4 \cdot 4\pi R_2^2$ and

$$\frac{R_2}{R_1} = \left(\frac{T_1}{T_2}\right)^2 = \left(\frac{5778}{4265}\right)^2 = 1.8.$$

The Sun expands to about 1.8 times its present radius. It is cooler at the surface but bigger, and the two effects cancel because luminosity depends on $T^4 R^2$.

4 The photoelectric effect

Silver has $\phi = 4.3 \text{ eV} = 4.3 \times 1.60 \times 10^{-19} = 6.88 \times 10^{-19} \text{ J}$.

(i) Maximum wavelength for emission

A photon needs at least enough energy to free an electron, so the threshold is $hf = \phi$. Since $f = c/\lambda$, less energy means longer wavelength, so the minimum energy gives a maximum wavelength:

$$\frac{hc}{\lambda_{\max}} = \phi \quad \implies \quad \lambda_{\max} = \frac{hc}{\phi} = \frac{6.63 \times 10^{-34} \times 2.998 \times 10^8}{6.88 \times 10^{-19}} = 289 \text{ nm}.$$

This is in the ultraviolet, so visible light will not eject electrons from silver no matter how bright it is. Classical wave theory cannot explain that, since it predicts a brighter beam should eventually supply enough energy.

(ii) Energy of the photoelectrons at 200 nm

The photon energy is

$$E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 2.998 \times 10^8}{200 \times 10^{-9}} = 9.94 \times 10^{-19} \text{ J},$$

so from $E_k = hf - \phi$,

$$E_k = 9.94 \times 10^{-19} - 6.88 \times 10^{-19} = 3.06 \times 10^{-19} \text{ J} = 1.91 \text{ eV}.$$

This is the *maximum* kinetic energy. Electrons starting below the surface lose some energy escaping, so in practice there is a spread of energies up to this value.

(iii) Photons absorbed per second

The silver receives $1.00 \times 10^{-6} \text{ J}$ every second and each photon carries $9.94 \times 10^{-19} \text{ J}$, so

$$N = \frac{P}{E} = \frac{1.00 \times 10^{-6}}{9.94 \times 10^{-19}} = 1.01 \times 10^{12} / \text{s}.$$

(iv) Photocurrent

One electron per photon means N electrons per second, each carrying charge e :

$$I = Ne = 1.01 \times 10^{12} \times 1.60 \times 10^{-19} = 1.61 \times 10^{-7} \text{ A} = 0.161 \mu\text{A}.$$

A real photocathode has a quantum efficiency well below one, so the measured current would be smaller.

(v) Stopping voltage

The current stops when the retarding potential is just enough to turn back the fastest electrons, so $eV_s = E_k$ and

$$V_s = \frac{E_k}{e} = \frac{3.06 \times 10^{-19}}{1.60 \times 10^{-19}} = 1.91 \text{ V}.$$

The stopping voltage in volts equals the kinetic energy in electronvolts, which follows directly from how the electronvolt is defined. It is also why the stopping voltage is such a convenient measurement: it gives the electron energy without needing to measure any speeds.

Model check. Silver is one of the metals in my Task 4 model, which plots $V_s = (hf - \phi)/e$ against frequency. The threshold from part (i) is where the silver line crosses the horizontal axis, at $f_0 = \phi/h = 1.04 \times 10^{15} \text{ Hz}$.

On its own that is not much of a check, since setting $V_s = 0$ in the model just recovers $hf = \phi$, which is the equation I solved by hand. A better test is to work backwards from the plotted line. Fitting a straight line to it gives

$$\text{gradient} = 4.1438 \times 10^{-15} \text{ V/Hz}, \quad \text{intercept} = -4.300 \text{ V},$$

and since $V_s = (h/e)f - \phi/e$, the gradient should be h/e and the intercept $-\phi/e$. Multiplying the gradient by e returns $h = 6.630 \times 10^{-34} \text{ Js}$ and the intercept returns $\phi = 4.300 \text{ eV}$, both exactly the values I put in.

This is worth doing because it is how the experiment is used in practice. Millikan measured h this way in 1916: you cannot weigh a photon, but you can measure stopping voltage against frequency and read Planck's constant off the gradient. The fact that the gradient is h/e for every

metal, so all the lines come out parallel, is what makes the measurement independent of which metal you happen to use.

Below the threshold my code returns NaN rather than a negative voltage, so the line stops instead of continuing into a region where no electrons are emitted at all.

5 Electron acceleration and diffraction

(i) Final velocity

All the electrical work done on the electron becomes kinetic energy, so

$$eV = \frac{1}{2}mv^2 \quad \implies \quad v = \sqrt{\frac{2eV}{m}}.$$

(ii) Condition for the classical treatment

Setting $v < 0.01c$ and squaring,

$$\frac{2eV}{m} < (0.01c)^2 = 10^{-4}c^2,$$

so

$$V < \frac{10^{-4}mc^2}{2e} = \frac{10^{-4} \times 9.109 \times 10^{-31} \times (2.998 \times 10^8)^2}{2 \times 1.60 \times 10^{-19}} = 25.6 \text{ V}.$$

Below this the relativistic factor $\gamma = (1 - v^2/c^2)^{-1/2}$ differs from 1 by less than 0.005%, so relativistic effects can be ignored and the classical expressions for momentum and kinetic energy used instead.

(iii) de Broglie wavelength

From the de Broglie relation $\lambda = h/p$. In the classical limit the kinetic energy is $K = p^2/2m$, so $p = \sqrt{2mK}$, and all the kinetic energy came from the accelerating voltage so $K = eV$. Therefore

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2meV}}.$$

This is the expression my Task 6 model uses to convert accelerating voltage into wavelength, so deriving it here justifies an assumption the code makes rather than just asserting it.

(iv) Intensity against angle

The pattern is a bright central spot with concentric rings around it, so a plot of intensity against angle shows a large central peak with smaller secondary maxima either side, falling away as the angle increases.

This happens because the electrons scatter off successive layers of carbon atoms. Waves reflected from adjacent layers travel different distances, and where that path difference is a whole number of wavelengths they arrive in phase and interfere constructively, giving a peak. Everywhere else they partly cancel.

The point is that interference is something only waves do. Getting a diffraction pattern from a beam of electrons means electrons must have a wave nature as well as behaving like particles, which is what the de Broglie relation says. This is direct experimental evidence for wave-particle duality.

(v) Accelerating voltage for a peak at 10.0°

The Bragg condition for constructive interference from layers a distance d apart is $2d \sin \theta = n\lambda$. For the first order peak $n = 1$:

$$\lambda = 2d \sin \theta = 2 \times 1.23 \times 10^{-10} \times \sin 10.0^\circ = 4.27 \times 10^{-11} \text{ m.}$$

Rearranging the result of part (iii) for V ,

$$\lambda = \frac{h}{\sqrt{2meV}} \quad \Rightarrow \quad V = \frac{h^2}{2me\lambda^2},$$

and substituting,

$$V = \frac{(6.63 \times 10^{-34})^2}{2 \times 9.109 \times 10^{-31} \times 1.60 \times 10^{-19} \times (4.27 \times 10^{-11})^2} = 828 \text{ V.}$$

This exceeds the 25.6 V limit from part (ii). At 828 V the electron reaches $v = 0.057c$, so the classical treatment is being used slightly outside the range where part (ii) strictly justifies it. The correction is second order in v/c and comes to about 0.2 %, small enough not to matter here, but it is an approximation rather than an exact result.

(vi) The remaining peaks

Higher orders satisfy the same condition with $n = 2, 3, \dots$, so

$$\sin \theta_n = \frac{n\lambda}{2d} = n \sin 10.0^\circ = 0.1736 n.$$

n	$\sin \theta_n$	θ_n
1	0.174	10.0°
2	0.347	20.3°
3	0.521	31.4°
4	0.695	44.0°
5	0.868	60.3°
6	1.042	none

There are exactly five peaks. At $n = 6$ the Bragg condition would need $\sin \theta > 1$, which no angle satisfies, so the series stops.

The peaks are evenly spaced in $\sin \theta$ but not in angle, because $\arcsin$ stretches the spacing as it approaches 90° . The gap from $n = 1$ to $n = 2$ is 10.3° , but from $n = 4$ to $n = 5$ it is 16.3° . A sketch should therefore show the outer peaks spreading further apart rather than a set of equally spaced lines.

Model check. My Task 6 model runs this calculation in the opposite direction, taking a voltage and returning the ring radius on the phosphor screen. Putting 828 V into it gives a de Broglie wavelength of 42.7 pm and a Bragg angle of 10.0° , recovering the value the question started from.

The model also plots $\sin(\phi/2)$ against $1/\sqrt{V}$, which is a straight line because part (iii) gives $\lambda \propto 1/\sqrt{V}$ and the Bragg condition makes $\sin(\phi/2)$ proportional to λ . Taking the gradient k of that line and rearranging,

$$d = \frac{h}{2k\sqrt{2m_e e}},$$

which returns 0.1230 nm and 0.2130 nm for the two spacings I put in. Recovering the input values confirms the linearisation is right. This matters because it is how the spacing would actually be measured: you cannot see the atomic layers directly, but you can measure ring radii against voltage and get d from the gradient.

6 Hydrogen spectral series

For a transition from level n down to level m , the Balmer formula with $Z = 1$ gives

$$\lambda_{nm} = 91.13 \text{ nm} \left(\frac{1}{m^2} - \frac{1}{n^2} \right)^{-1}.$$

Lyman line, $n = 3 \rightarrow m = 1$

$$\lambda = \frac{91.13}{\frac{1}{1} - \frac{1}{9}} = \frac{91.13}{8/9} = 102.5 \text{ nm}.$$

Far ultraviolet. Lyman lines all end on the ground state, so they involve the biggest energy gaps and therefore the shortest wavelengths.

Balmer line, $n = 4 \rightarrow m = 2$

$$\lambda = \frac{91.13}{\frac{1}{4} - \frac{1}{16}} = \frac{91.13}{3/16} = 486.0 \text{ nm}.$$

Visible, in the blue-green. This is the $H\beta$ line. Balmer is the only one of the five series with lines in the visible range, which is why it was found first.

Pfund line, $n = 7 \rightarrow m = 5$

$$\lambda = \frac{91.13}{\frac{1}{25} - \frac{1}{49}} = \frac{91.13}{24/1225} = 4651 \text{ nm}.$$

Mid infrared, about ten times longer than the Balmer line. Pfund lines end on $m = 5$, where the levels are already close together, so the energy gaps are small and the wavelengths long.

Comment

These three lines span a factor of more than forty five in wavelength even though they all come from the same atom and the same formula. That is the $1/n^2$ spacing of the energy levels: transitions ending on low m have large gaps and short wavelengths, while those ending on high m have gaps that have already shrunk a lot.

Model check. My Task 5 model works from the level energies rather than the 91.13 nm formula, so it is an independent check. It gives 102.70 nm, 486.89 nm and 4659.71 nm, agreeing to within 0.2 %.

The small offset is not rounding. My model uses the reduced mass $\mu = m_e M / (m_e + M)$ rather than m_e alone, which allows for the nucleus recoiling instead of being infinitely heavy. Since μ is about 0.05 % smaller than m_e , every level is slightly shallower and every wavelength slightly longer, which is the direction the difference actually goes.

The model also takes Z as a parameter. Increasing it moves every line to shorter wavelength as Z^2 : at $Z = 2$ the Lyman limit drops from 91.3 nm to 22.8 nm, a factor of exactly four.
